

- 11** The enthalpy change of formation of potassium bromide, KBr, can be calculated using a Born-Haber cycle.

The enthalpy changes related to potassium and bromine are shown in the table.

	enthalpy change /kJ mol ⁻¹
$\text{K(s)} \rightarrow \text{K(g)}$	+90
$\text{Br}_2\text{(g)} \rightarrow 2\text{Br(g)}$	+193
$\text{K(g)} \rightarrow \text{K}^+\text{(g)} + \text{e}^-$	+418
$\text{Br(g)} + \text{e}^- \rightarrow \text{Br}^-\text{(g)}$	-325
$\text{K}^+\text{(g)} + \text{Br}^-\text{(g)} \rightarrow \text{KBr(s)}$	-678

What is the enthalpy change of formation of KBr?

- A** -302 kJ mol⁻¹
B -399 kJ mol⁻¹
C -958 kJ mol⁻¹
D -1054 kJ mol⁻¹